

ORCHESTRA-OS: Simulation-Driven Architectural Discovery and Design Principles for Predictive, Hierarchical, Signal-Coordinated Process Scheduling

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Abstract—Signal-driven scheduling architectures coordinate large populations of operating-system processes through a shared, predictively extrapolated, cryptographically verified hardware signal rather than through syscall-mediated centralized arbitration. This paper presents ORCHESTRA-OS, a hierarchical signal-coordinated scheduling architecture spanning core, node, and cluster tiers, and reports a discrete-event, userspace simulation study—spanning seven iterative rounds of validation—built expressly to test the architecture’s load-bearing claims prior to kernel implementation. We show that a naive implementation of the architecture’s tri-factor coordination index $Q(t) = S_1 S_2 S_3$ contains a metric blind spot that allows a trivially synchronized population to score a perfect coordination value while exhibiting the exact thundering-herd behavior the architecture is meant to prevent; we introduce a fourth factor, S_4 , that closes this gap. We show that a single-actuator PID feedback controller saturates without effect when the population’s coordination deficit is not caused by the actuator’s target quantity, and replace it with a multi-actuator controller exhibiting two-timescale separation from the underlying multi-agent reinforcement learning process. We report a failed attempt at online adaptive Kalman filtering (mean squared error $7\times$ worse than a naive baseline), diagnose its cause as a classical noise-identifiability problem, and replace it with an offline-calibrated estimator. We apply global-games equilibrium-selection theory to size an anti-synchronization noise term from first principles, and difference-reward credit assignment to give learning agents a direct local incentive toward population-level coordination. Across all rounds, the simulated coordination index rises from 0.40 to 0.868, ultimately exceeding a structurally advantaged reactive baseline (0.861). Cryptographic signal integrity is confirmed by 100% detection of all injected tamper events across the simulation. Finally, we distill the study into ten architecture-independent design principles for signal-driven operating systems, applicable to distributed schedulers, hypervisors, container orchestration, and edge resource management.

Index Terms—Operating system scheduling, parallel and distributed systems, multi-agent reinforcement learning, coordination games, Kalman filtering, global games, difference rewards, stochastic approximation, simulation-based validation.



1 INTRODUCTION

MODERN process schedulers coordinate CPU-bound and I/O-bound work through per-core run queues with limited cross-core signaling, typically restricted to load-balancing heuristics evaluated at coarse intervals. Every scheduling decision in mainstream centralized schedulers, such as the Completely Fair Scheduler [6], requires a system call and centralized kernel arbitration, introducing context-switch overhead that scales poorly with core count. Rigid real-time schedulers (SCHED_FIFO / SCHED_RR) provide deterministic ordering but cannot adapt to fluctuating thermal, memory, or congestion conditions. Existing shared-memory kernel-to-user signaling mechanisms (vDSO-style counters, lightweight hint pages) expose limited, single-purpose data rather than a structured, multi-

dimensional system-health vector, and carry no cryptographic integrity protection. Moreover, decades of work on dynamic load balancing in parallel and distributed systems [1] has established that populations of clients reacting to shared—and especially stale—load state exhibit a well-documented “herd effect,” in which apparently underutilized resources are simultaneously flooded by all clients [3], [2]; the classical remedies are randomization of client choices [4] and age-aware interpretation of the shared state [3].

ORCHESTRA-OS, for which a patent application has been submitted by the present authors, addresses these gaps with a fundamentally different coordination model: an explicit, cryptographically signed, predictively extrapolated *signal bus* organized hierarchically across core, node, and cluster tiers. A small set of system-wide hardware indicators (CPU utilization, thermal headroom, memory pressure) is sampled at fixed intervals, forecast one step ahead, signed, and exposed read-only via memory mapping to every scheduled process—no system call is required to read it. Each process runs a local *Adaptive Response Function* that maps the signal to one of a small set of scheduling actions (run, sleep, migrate, throttle, yield), and a per-tier *Coordination Index* $Q(t)$ quantifies how well the population’s

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collective behavior tracks the signal’s intent. Section 2 describes the architecture in full.

Architectures of this kind make three load-bearing claims prior to any kernel implementation: (i) that the signal can be forecast accurately and verified cheaply enough to run on every scheduler tick; (ii) that the coordination index is a faithful measure of the population’s actual behavior, including pathological synchronized behavior; and (iii) that a population of independently adapting processes can be steered toward high coordination without direct inter-process communication. None of these claims is self-evidently true, and each is more cheaply falsified in a userspace simulator than in a kernel patch. Indeed, the submitted patent application itself identifies software simulation of the signal bus, predictive layer, and adaptive response function as the necessary next step prior to kernel-level implementation; this paper is that step.

More broadly, the paper illustrates how simulation can function as a scientific instrument for architectural discovery rather than merely a stage of implementation validation.

We document a Phase 1 (pre-implementation) simulation study built expressly to test these three claims, structured as a sequence of seven rounds, each triggered by a concrete failure the simulation itself surfaced. Our contributions are:

- We present the ORCHESTRA-OS architecture—predictive extrapolation, cryptographic signal integrity, tiered core/node/cluster signal hierarchy, adaptive per-process response, hybrid real-time safety bypass, and closed-loop coordination control—as a coherent design for decentralized scheduling in parallel and distributed systems (Section 2).
- We identify and repair a genuine *metric blind spot* in the originally disclosed tri-factor coordination index: a synchronized population can score a perfect value while exhibiting thundering-herd behavior, because the original index is a set of single-tick population snapshots with no temporal-rate term (Section 5.2).
- We identify and repair a *controller–target mismatch*: a single-lever PID controller saturates without improving the coordination index when the deficit is not caused by the quantity the lever controls, and we replace it with a multi-actuator controller with explicit two-timescale separation from the underlying learning process, as required by stochastic approximation theory for joint convergence guarantees to hold (Section 5.3).
- We report a *negative result*: an online-adaptive Kalman filter for the predictive layer performs substantially worse than a naive fixed-parameter baseline; we diagnose the failure as a classical noise-covariance identifiability problem and replace it with an offline-calibrated estimator (Section 5.7).
- We apply *global-games* equilibrium-selection theory to derive, rather than hand-tune, the magnitude of an anti-synchronization noise term, and validate the derived value against ground truth (Section 5.8).
- We apply *difference-reward* multi-agent credit assignment to give individually learning agents a direct incentive toward population-level coordination, closing

ing a gap where all coordination pressure had previously been externally imposed (Section 5.9).

- We report a multi-seed sensitivity analysis over the two remaining free parameters introduced by the above fixes, distinguishing genuine effects from single-run noise (Section 6.4).
- We distill the study into *ten architecture-independent design principles* for signal-driven operating system design, each stated formally, grounded in the paper’s empirical evidence, and generalized beyond the ORCHESTRA-OS context (Section 7).

2 THE ORCHESTRA-OS ARCHITECTURE

This section summarizes the architecture as specified in the submitted patent application. ORCHESTRA-OS is built around four core technical elements operating in a closed loop executed once per scheduling tick.

2.1 Predictive Extrapolation Layer

At a fixed tick (illustratively every 10 ms, tunable by the feedback controller of Section 2.4), the kernel samples raw hardware counters—per-core CPU utilization, free memory pages, thermal sensor readings, cache hit/miss ratios, disk queue depth, and network buffer occupancy. A lightweight forecasting function, sized to execute within the kernel’s tick budget, analyzes the recent history of sampled values and produces a short-horizon forecast $G(t + \Delta)$, which is combined with the raw sample into the signal frame. Predictive extrapolation is the architecture’s primary defense against reactive lag and synchronized “herd” oscillation, in which many processes react identically to the same stale signal.

2.2 Signal Integrity Module and Tiered Mapping

The kernel writes the raw values and the forecast into a structured, fixed-size *Global Signal Vector* for each tier—core, node, and cluster—and signs each resulting frame with a rotating-key HMAC together with a monotonically increasing sequence number. Each signed frame is mapped read-only into the address space of every entitled process via `mmap`; once mapped, a process reads the signal as it would any variable in memory, with no system call. A process on a given core reads $G_{\text{local}}(t)$; node- and cluster-tier frames aggregate the same quantities at coarser granularity and cadence, allowing the identical coordination principle to scale from a single embedded core to multi-core, NUMA, and distributed cluster deployments without redesign. Before trusting a frame, a reading process verifies the HMAC signature and sequence number, rejecting stale, duplicated, or corrupted frames—closing a spoofing/tampering gap present in prior shared-memory hint mechanisms.

2.3 Adaptive Response Function and Hybrid Safety Layer

At each scheduling opportunity, a process’s Adaptive Response Function—a lightweight policy that adapts over time via reinforcement-learning-style reward feedback—reads the verified signal and selects one of five actions: RUN, SLEEP, MIGRATE, THROTTLE, or YIELD. Before any self-scheduling decision takes effect, the Hybrid Safety Layer

checks whether the requesting task carries a hard real-time flag (SCHED_FIFO or SCHED_RR); if so, the task bypasses the signal-driven path entirely and is routed through the conventional deterministic scheduler, preserving hard real-time guarantees in mixed-criticality deployments.

2.4 Coordination Index and Closed-Loop Feedback Controller

The kernel continuously computes a per-tier Coordination Index. As originally disclosed, the index is the product of three normalized factors,

$$Q(t) = S_1(t) S_2(t) S_3(t), \quad Q(t) \in [0, 1], \quad (1)$$

where the *freshness factor*

$$S_1(t) = e^{-\alpha \Delta t_{\text{sample}}} e^{-\beta \Delta t_{\text{forecast}}} \quad (2)$$

degrades with the age of the most recently verified frame and the divergence between forecast and subsequently observed values; the *compliance factor*

$$S_2(t) = N_{\text{compliant}}(t) / N_{\text{eligible}}(t) \quad (3)$$

is the fraction of non-exempt signal-path processes whose selected action matches the directive indicated by the verified signal frame; and the *coherence factor*

$$S_3(t) = 1 - \sigma_{\text{actions}}^2(t) / \sigma_{\text{max}}^2 \quad (4)$$

measures the consistency of responses among similarly situated processes, normalized against the variance of a fully divergent action distribution. A Closed-Loop Feedback Controller (specified in the disclosure as PID-style) monitors the trend of Q at each tier and continuously, proportionally retunes the sampling tick interval, safety-layer thresholds, and prediction horizon—a graduated mechanism in place of a binary reassertion of centralized control.

At the time of disclosure the architecture was at the conceptual and design stage; the simulation study that follows is the validation step the disclosure identifies as prerequisite to kernel implementation. As Sections 5 and 6 show, the simulation both validates the architecture’s core mechanisms and materially revises two elements of the specification above: the form of the coordination index (1) and the design of the feedback controller.

3 RELATED WORK

Load balancing with shared and stale state. Dynamic load balancing on parallel machines has been studied extensively, with classical strategy taxonomies established by Willebeek-LeMair and Reeves [1]. Two results from this literature bear directly on the present work. First, Mitzenmacher [2] and Dahlin [3] showed that naively sending work to the apparently least-loaded server behaves pathologically when the shared load information is stale—the “herd effect,” in which all clients simultaneously flood the same apparently idle resource—and that remedies include randomization and interpreting load information as a function of its age. The power-of-two-choices result [4] formalizes how a small amount of per-client randomness dramatically improves balance; ORCHESTRA’s per-agent perceptual jitter (Section 5.8) serves the analogous symmetry-breaking role,

but is sized from equilibrium-selection theory rather than chosen as a protocol constant, and the S_4 stability factor (Section 5.2) makes herd episodes directly measurable in the coordination index. Second, adaptivity in load balancing has precedent: Zomaya and Teh [5] studied learning-based (genetic-algorithm) dynamic load balancing, observing both the promise and the tuning burden of adaptive policies—a burden this paper addresses by deriving, rather than tuning, the parameters its adaptive mechanisms introduce.

Exponential smoothing and Kalman filtering. Simple exponential smoothing and its trend-aware extension [7] are widely used for low-overhead forecasting. For a local-level (random-walk-plus-noise) generative model, the steady-state Kalman gain [8] coincides with the optimal exponential smoothing constant, making the choice of smoothing constant equivalent to a noise-covariance estimation problem. Online estimation of Kalman filter noise covariances from the innovation sequence has a long history [9]; a central result of that literature, confirmed empirically in this work, is that process and observation noise are not jointly identifiable from lag-zero innovation statistics alone.

Stochastic approximation and two-timescale systems. The convergence of tabular Q-learning [16] relies on Robbins–Monro step-size conditions under a stationary Markov decision process. When an outer control loop modifies the reward function an agent is learning against, the coupled system is no longer stationary; two-timescale stochastic approximation theory [10], [11] gives conditions—principally, that the outer loop’s updates vanish faster than the inner loop’s—under which the coupled system nonetheless converges to a solution of an associated pair of ordinary differential equations.

Global games. Coordination games with a public signal often admit multiple equilibria, including inefficient, synchronized ones. Carlsson and van Damme [12] and Morris and Shin [13] show that perturbing a public signal with private, idiosyncratic agent-specific noise eliminates equilibrium multiplicity in the small-noise limit, selecting a unique threshold equilibrium. This result is used in economics primarily to explain phenomena such as bank runs and currency-attack coordination; we apply it here to justify and size an anti-synchronization perturbation in a scheduling context.

Multi-agent credit assignment. When many agents share an environment and a global reward signal, an individual agent’s learning signal is diluted by the actions of all other agents. Difference rewards, also called Aristocrat utility [14], address this by crediting each agent with the counterfactual effect of its own action on a global utility function, holding all other agents’ actions fixed.

4 SIMULATION MODEL

We simulate a single scheduling tier (core-local aggregation) of the architecture of Section 2: $N = 40$ processes over $T = 3000$ discrete ticks. Four processes are exempt real-time processes that bypass the signal-driven path entirely, per the Hybrid Safety Layer.

4.1 Signal Bus

At each tick, a hardware model generates CPU utilization $u_t \in [0, 1]$ as a slow sinusoidal baseline plus Poisson-arrival

load spikes and Gaussian sensor noise; thermal state follows a first-order lag of u_t with per-tick smoothing coefficient 0.05 (time constant $\tau \approx 20$ ticks), representing the thermal inertia of a typical processor package. A signal frame containing u_t , thermal state, memory pressure, and a one-step-ahead forecast is signed with a rotating-key HMAC and a monotonic sequence number. A verifying process rejects frames that fail signature verification or whose sequence number indicates staleness beyond a fixed window—a direct instantiation of the Signal Integrity Module.

4.2 Coordination Index

Let $D(u_t, \theta_t) \in \mathcal{A}$ denote the *directive*—the action a compliant process should take given the true signal state, where $\mathcal{A} = \{\text{run, sleep, migrate, throttle, yield}\}$. We instantiate the disclosed factors as four normalized quantities in $[0, 1]$:

- S_1 (signal fidelity): forecast accuracy combined with a staleness decay, instantiating the disclosed freshness factor.
- S_2 (directive compliance): $N_{\text{compliant}}(t)/N_{\text{eligible}}(t)$, the fraction of non-exempt signal-path processes whose selected action matches the directive indicated by the verified signal frame, instantiating the disclosed compliance factor.
- S_3 (action coherence): one minus the normalized entropy of the population’s action distribution.
- S_4 (temporal stability, introduced in Section 5.2): one minus the fraction of non-exempt processes whose action changed since the previous tick.

The coordination index is their geometric mean, $Q(t) = (S_1 S_2 S_3 S_4)^{1/4}$ (see Section 5.1 for why this is a geometric mean rather than the raw product of (1)).

4.3 Adaptive Response Function

Each non-exempt process is a tabular Q-learning agent [16] over a discretized (cpu_bucket, thermal_bucket) state space, selecting an action via ϵ -greedy exploration and updating via the standard temporal-difference rule. We describe the discretization, reward function, and exploration schedule as originally implemented, and their subsequent repair, in Sections 5.4–5.6.

4.4 Baseline

A reactive baseline applies the directive function $D(u_t, \theta_t)$ directly as every non-exempt process’s action, with no forecasting, no cryptographic verification, and no learning. This baseline is deliberately naive: it represents a scheduler with no coordination machinery, and—critically—computes the ground-truth directive verbatim rather than inferring it, giving it an artificially high ceiling on S_2 that no learned policy can be expected to exceed by construction.

5 DESIGN ISSUES FOUND AND CORRECTED

5.1 Aggregation Operator: Product versus Geometric Mean

A product of n sub-unity terms is not a mean: it lacks the normalizing $1/n$ root of a proper power mean, and therefore

shrinks mechanically whenever a new factor is introduced, independent of whether underlying behavior changed. We verified this directly: four factors valued at approximately 0.88 each multiply to approximately 0.60, a 28-point drop from their arithmetic average. This exactly explains an artifact observed when S_4 was first introduced (Section 5.2): the reported coordination index fell even though no component’s value had worsened. We replace the raw product of (1) with the geometric mean, $Q(t) = (S_1 S_2 \cdots S_n)^{1/n}$, which retains the same “any factor near zero collapses the score” AND-like semantics (it remains the $p \rightarrow 0$ member of the power-mean family) while remaining comparable in magnitude across revisions of the index with different numbers of factors.

5.2 The S_4 Metric Blind Spot

The originally disclosed index (1) scores only single-tick population snapshots. A population in which every process applies an identical deterministic rule to an identical instantaneous signal trivially maximizes both S_2 and S_3 —including at the instant the entire population flips its collective action simultaneously in response to a threshold crossing, which is precisely the thundering-herd behavior the architecture is intended to prevent and which has long been documented in load balancing over shared state [3], [2]. We confirmed this directly: the baseline defined in Section 4 scored a perfect $Q(t) = 1.0$ under the original index while visibly exhibiting synchronized mass-switching. We introduce S_4 , defined above, to penalize this directly; under the corrected, S_4 -aware index, the baseline’s steady-state score falls from 1.00 to 0.86 (Table 1), correctly reflecting its periodic stampede episodes. We acknowledge an inherent tradeoff: S_4 penalizes any action change, including coordinated collective action that correctly tracks a signal threshold crossing. Temporal stability and signal responsiveness are thus in tension by design; the controller’s role is to balance these via the jitter and switching-penalty levers, rather than to eliminate either.

5.3 Controller–Target Mismatch and Two-Timescale Repair

The disclosed design closes the loop with a PID-style controller retuning a single parameter—the predictive layer’s smoothing constant—toward $Q(t) \rightarrow 1$. We found that this controller saturates its actuator at the parameter’s ceiling and remains pinned there for the entire simulation, because S_1 (the quantity the actuator affects) was already high (≈ 0.92); the actual deficit was in S_2 and S_3 (population-level policy divergence), which the smoothing constant has no causal effect on. We replace the single-lever controller with a three-actuator design that decomposes the rolling-average deficit across S_3 and S_4 and selects the matching lever: a per-agent perceptual-jitter magnitude (targeting S_4), a shared switching penalty in the agents’ reward function (targeting S_3/S_4), and a periodic Q-table consensus blend across agents (targeting S_3 ; added after observing that jitter and switching penalty alone can only suppress disagreement, not manufacture agreement between independently converged policies).

Because the outer controller mutates the agents' reward function (via the switching penalty), the coupled system violates the stationarity assumption underlying standard tabular Q-learning convergence proofs. Two-timescale stochastic approximation theory [10], [11] restores a convergence guarantee provided the controller's own updates vanish relative to the learners': we enforce this by updating the controller only once every 20 ticks, with a step size that decays proportionally to $k^{-1/2}$ on the controller's own update counter k , rather than every tick at constant magnitude. Formally, the learners use a constant temporal-difference step size $\alpha = 0.2$ —the standard constant-stepsizes regime, appropriate for tracking in a slowly varying environment—while each controller actuator uses $\beta_k = \beta_0(1 + k)^{-1/2}$ with $\beta_0 \in \{0.01, 0.02\}$ applied on the controller's own update counter; since $\beta_k \rightarrow 0$ while α remains constant, the timescale-separation ratio $\beta_k/\alpha \rightarrow 0$ required by the two-timescale framework [10], [11] is satisfied.

5.4 Reward–Directive Contradiction

Inspecting the originally implemented reward function against the directive function it is scored against revealed a direct contradiction: the reward rewarded the RUN action for $u_t \in [0.3, 0.7]$, while the directive specifies MIGRATE for $u_t > 0.6$. Agents were therefore being correctly trained toward a target that actively disagreed with the metric used to score them—a bug in the reward specification, not a hard exploration–exploitation tradeoff. We replaced the reward with one driven directly by directive match (+0.6 if the chosen action equals the directive, −0.6 otherwise), retaining an independent thermal-safety term as defense in depth. This single change raised mean steady-state S_2 from 0.57 to 0.79 (Table 1).

5.5 State Discretization Misalignment

A uniform four-bin discretization of u_t was found to place genuinely different correct directives within the same bucket—for example, both $u_t = 0.60$ (RUN) and $u_t = 0.65$ (MIGRATE) fell in the same bucket. Because a tabular Q-learner can assign only one preferred action per state, this constitutes a hard ceiling on achievable compliance independent of training duration, not a noise or tuning problem. We re-aligned bucket boundaries exactly to the directive function's own decision thresholds, raising mean steady-state S_2 from 0.79 to 0.90.

5.6 Exploration Annealing

A fixed exploration rate of $\varepsilon = 0.1$ caps achievable compliance at approximately $1 - \varepsilon + \varepsilon/|\mathcal{A}| \approx 0.92$ regardless of policy quality. We introduce standard exponential annealing, $\varepsilon_k \leftarrow \max(\varepsilon_{\min}, \varepsilon_k \cdot \rho)$, from $\varepsilon_0 = 0.3$ to $\varepsilon_{\min} = 0.02$. The decay rate is set to $\rho = 0.998$ per tick, reaching the floor $\varepsilon_{\min} = 0.02$ by approximately tick 1350 of 3000.

5.7 Predictor: A Failed Online-Adaptive Kalman Filter

The exponential-smoothing predictor's smoothing constant was originally either hand-set or retuned by the PID controller described in Section 5.3—a controller shown in that

section to be decoupled from the quantity it was intended to affect. Observing that a local-level model's steady-state Kalman gain is analytically determined by the ratio of process to observation noise, we first attempted *online* estimation of this ratio from the innovation sequence. This attempt failed: measured against a 3000-tick evaluation trace, the online-adaptive filter's mean squared forecast error was 0.0286, roughly seven times worse than a naive fixed-parameter exponential smoother (MSE 0.0041–0.0057; Fig. 2). Tracing internal filter state identified the cause as the classical noise-identifiability problem described by Mehra [9]: process and observation noise variances are not separable from lag-zero innovation variance alone. When a genuine load spike arrived while the filter's estimated uncertainty was small, the entire innovation was misattributed to observation noise, collapsing the Kalman gain and causing the filter to distrust real data for over 100 subsequent ticks. We report this negative result rather than omit it, since the failure mode is instructive and not specific to this codebase.

We replace online estimation with *offline calibration*: the single free parameter $\lambda = Q_{\text{proc}}/R_{\text{obs}}$ (only the ratio affects steady-state filter behavior for this model class) is fit by a one-dimensional grid search minimizing one-step-ahead mean squared error on a held-out calibration trace generated with an independent random seed, avoiding any look-ahead into the evaluation trace. The calibrated filter achieves MSE 0.00404, matching the best hand-tuned fixed smoothing constant found in this study (0.00406) to within noise, while being derived from a procedure that generalizes to other hardware traces rather than a value discovered by an unrelated controller saturating at its bound.

5.8 Global-Games-Derived Jitter Sizing

The population studied here is an instance of a shared-public-signal coordination game: many agents observe a common signal and select an action, admitting a synchronized equilibrium as one possibility among several. The load-balancing literature addresses the analogous pathology through per-client randomization, most famously the power of two choices [4], but the amount of randomness is there a protocol design constant. Global-games theory [12], [13] shows that perturbing each agent's observation of the common signal with nonzero private noise eliminates this equilibrium multiplicity in the small-noise limit, and further implies how much noise suffices. Rather than allow the outer controller (Section 5.3) to discover a useful jitter magnitude purely through slow, reactive feedback from zero, we estimate the signal's intrinsic observation-noise standard deviation directly from the calibration trace using a robust (outlier-resistant) estimator based on the median absolute deviation of first differences [15], chosen specifically for robustness against the load-spike outliers that caused the failure in Section 5.7. Against a hardware model with a known, exact injected noise standard deviation of 0.030, the estimator recovered 0.0297 from a 500-tick calibration trace. The controller's jitter floor is set to $c\sigma_{\text{obs}}$ with $c = 1.5$ and initialized at this value at tick zero, rather than at zero. We emphasize that global-games theory is applied here as a principled analogy providing theoretical guidance for the

TABLE 1
Steady-state metrics across all seven rounds (seed 42).

Metric	R1	R2	R3	R4	R5	R6
Q_{ORCH}	0.40 ^a	0.32 ^a	0.47 ^a	0.66 ^a	0.86	0.868
Q_{base}	1.00 ^b	0.86	0.86	0.86	0.86	0.861
S_2	0.60	0.57	0.79	0.90	0.91	0.897
S_3	0.65	0.71	0.76	0.85	0.85	0.856
S_4	—	0.74	0.79	0.86	0.84	0.858

^aUses the raw-product formula (Section 5.1); not comparable in magnitude to R5–R6’s geometric mean. ^bReflects the pre- S_4 metric blind spot (Section 5.2).

form and scaling of the perturbation, rather than as a formal proof applicable to this specific multi-action, continuous-state setting.

5.9 Difference-Reward Credit Assignment

Prior to this fix, no individual agent’s reward reflected its own causal contribution to population-level coordination (S_3, S_4); all coordination pressure was imposed externally via jitter and consensus blending. We introduce a difference reward [14]: writing $G(\mathbf{a}) = (S_2(\mathbf{a})S_3(\mathbf{a})S_4(\mathbf{a}))^{1/3}$ for the population-level utility induced by the joint action vector $\mathbf{a}$, each non-exempt agent i receives

$$D_i = n (G(\mathbf{a}) - G(\mathbf{a}_{-i}, a_i^0)), \quad (5)$$

where a_i^0 is a fixed baseline action and n is the number of scored agents, used to rescale a marginal effect that would otherwise be $O(1/n)$ by construction. The difference reward is blended with the local directive-match reward at weight $w = 0.3$, keeping the unambiguous local signal dominant (justified empirically in Section 6.4).

6 EXPERIMENTAL RESULTS

All experiments use $N = 40$ processes (4 exempt), $T = 3000$ ticks, and a 500-tick warm-up exclusion before computing steady-state statistics, unless otherwise noted.

6.1 Round-by-Round Coordination Index

Figure 1 summarizes $Q(t)$ across all rounds for both systems. By the final round, ORCHESTRA’s coordination index (0.868) slightly exceeds the baseline’s (0.861), despite the baseline retaining a structural advantage: it computes the ground-truth directive directly rather than inferring it through a learned policy, and can lose points only to S_4 (temporal stability), never to S_2 or S_3 . Cryptographic signal integrity held throughout: all seven tamper events injected during the run (one every 401 ticks) were detected by HMAC verification and the corresponding frames rejected—a 100% detection rate—with the S_1 staleness decay engaging correctly on each rejection.

6.2 Predictor Comparison

Figure 2 compares one-step-ahead forecast MSE across all five predictor variants studied. The offline-calibrated Kalman filter matches the best hand-tuned exponential smoother while requiring no external retuning mechanism;

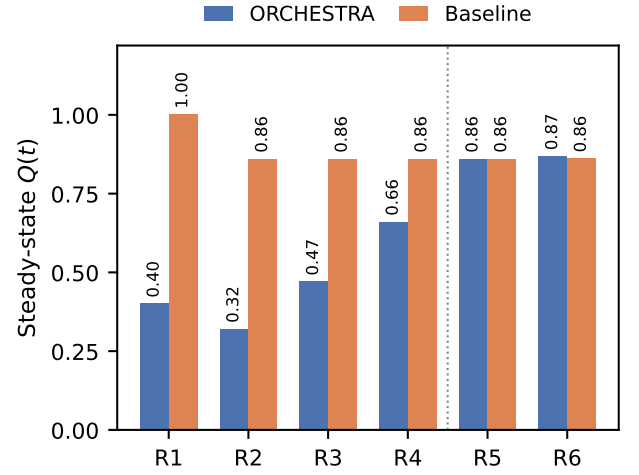


Fig. 1. Steady-state coordination index $Q(t)$ across all experimental rounds. Round key: R1 original specification; R2 S_4 and multi-actuator controller; R3 directive-driven reward; R4 aligned discretization and exploration annealing; R5 geometric-mean index, calibrated Kalman predictor, and two-timescale controller; R6 global-games jitter and difference rewards. Values left of the dotted line use the original raw-product formula; values right of it use the corrected geometric mean (Section 5.1) and are not directly comparable in magnitude to earlier rounds.

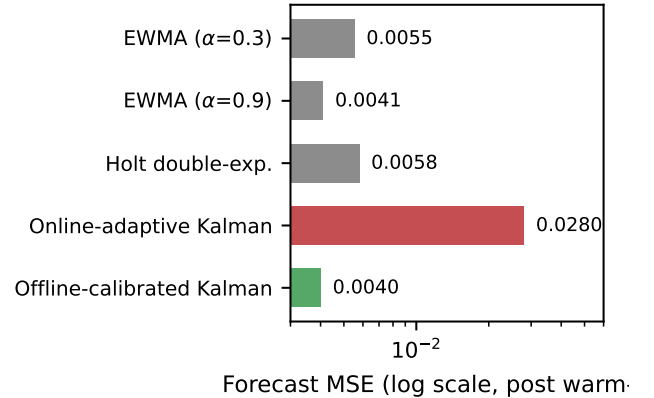


Fig. 2. One-step-ahead forecast MSE (log scale, post warm-up) across predictor variants. The online-adaptive Kalman filter (Section 5.7) is markedly worse than every fixed-parameter alternative; the offline-calibrated variant matches the best hand-tuned smoothing constant.

the online-adaptive Kalman filter is markedly worse than every fixed-parameter alternative, illustrating that “adaptive” is not synonymous with “better” when the underlying estimation problem is not identifiable from the available statistics.

6.3 Temporal Stability Comparison

Figure 3 shows S_4 for both systems over a representative 500-tick window. The baseline exhibits sharp, periodic collapses corresponding to population-wide synchronized switching; ORCHESTRA’s controller-and-jitter mechanism visibly suppresses the depth and duration of these episodes.

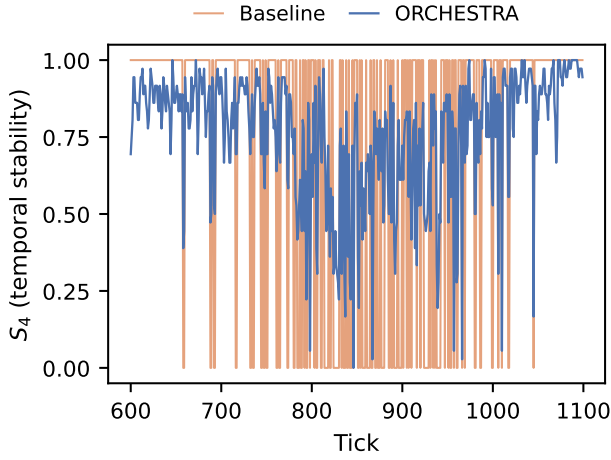


Fig. 3. Temporal stability (S_4) for baseline and ORCHESTRA over ticks 600–1100. The baseline’s periodic collapses correspond to population-wide synchronized action switching.

6.4 Sensitivity Analysis

The jitter multiplier c (Section 5.8) and difference-reward weight w (Section 5.9) were introduced without formal derivation of their exact numerical values. We conducted a multi-seed sensitivity sweep (5 seeds per setting for each one-dimensional sweep, 3 seeds per cell for a joint grid) to determine whether the chosen defaults ($c = 1.5$, $w = 0.3$) are defensible.

Figure 4(a) shows that steady-state $Q(t)$ as a function of c traces the theoretically expected tradeoff: S_2 falls and S_4 rises monotonically as jitter increases, since greater perceptual noise distorts an agent’s view of the true signal while more effectively desynchronizing threshold crossings. $Q(t)$, as a geometric mean, is maximized in an interior region rather than at either extreme; the chosen default lies within one combined standard deviation of the empirical maximum across the tested range $c \in [0, 5]$, and we found no statistical basis to revise it. Figure 4(b) shows the same qualitative tradeoff for w , with one additional finding: at $w = 1.0$ (reward driven entirely by the difference-reward term, with no local directive-match signal at all), the standard deviation of $Q(t)$ across seeds increases nearly fivefold (0.0296 versus ≈ 0.006 at every other tested value), indicating that removing the local reward signal entirely destabilizes learning outcomes across random seeds, not merely lowering their mean.

Table 2 and Fig. 5 report a joint grid over both parameters (3 seeds per cell). The default $w = 0.3$ is the best-performing setting at every tested jitter level, indicating the two mechanisms are complementary rather than substitutable within the useful range. However, the performance gap between $w = 0.3$ and $w = 0.7$ widens sharply as jitter increases (from 0.0017 at $c = 0$ to 0.0122 at $c = 3.0$), indicating a genuine interaction: elevating both parameters simultaneously compounds their respective noise contributions rather than acting independently.

TABLE 2
Joint sensitivity grid: mean steady-state $Q(t)$ (3 seeds per cell).

Team weight w	Jitter multiplier c		
	0.0	1.5	3.0
0.0	0.8630	0.8622	0.8561
0.3	0.8645	0.8660	0.8593
0.7	0.8628	0.8623	0.8471

7 IMPLICATIONS: PRINCIPLES FOR SIGNAL-DRIVEN OPERATING SYSTEM DESIGN

The simulation study reported here is not merely a validation exercise for one proposed architecture. Although the individual design corrections reported in Section 5 arose from ORCHESTRA-OS, repeated examination revealed that each reflected a more general systems-design principle rather than an implementation-specific repair. We therefore restate them below as architecture-independent design principles for signal-driven operating systems, applicable to any operating system that coordinates adaptive processes through a shared predictive signal. Each principle is stated as a formal proposition grounded in the empirical and theoretical findings of this paper, accompanied by the evidence base from which it is derived and its generalization beyond the specific ORCHESTRA-OS context. Table 3 summarizes the ten principles and their primary empirical evidence from this study.

7.1 Principle 1 — Coordination Metrics Must Encode Temporal Stability in Addition to Instantaneous Compliance

Formal statement. A coordination metric for a shared-signal scheduling architecture is adequate only if it is sensitive to temporal instability in the population’s action trajectory—specifically, to synchronized mass-switching across the process population—and not merely to the degree of single-tick agreement between individual process actions and a global directive.

Derivation from this study. The original tri-factor index (1) evaluated compliance (S_2) and action coherence (S_3) at each tick independently. Under this metric, a population in which every process applied an identical deterministic rule to the shared signal trivially scored $Q = 1$, including at the precise tick when the entire population switched action simultaneously in response to a threshold crossing. This is the thundering-herd failure mode—the class of behavior the architecture is designed to prevent—and the metric could not detect it. The introduction of the temporal stability factor S_4 reduced the reactive baseline’s score from 1.00 to 0.861, correctly penalizing its periodic mass-switching episodes while leaving a genuinely stable population unpenalized.

Generalization. Distributed schedulers, hypervisors, container orchestration systems, and edge-computing resource managers that expose a shared predictive state to many adaptive clients face structurally identical risks: a population of clients reacting uniformly to the same threshold crossing degrades the very resource the signal is meant to protect. Any coordination index for such systems should incorporate a rate-of-change term over the population’s

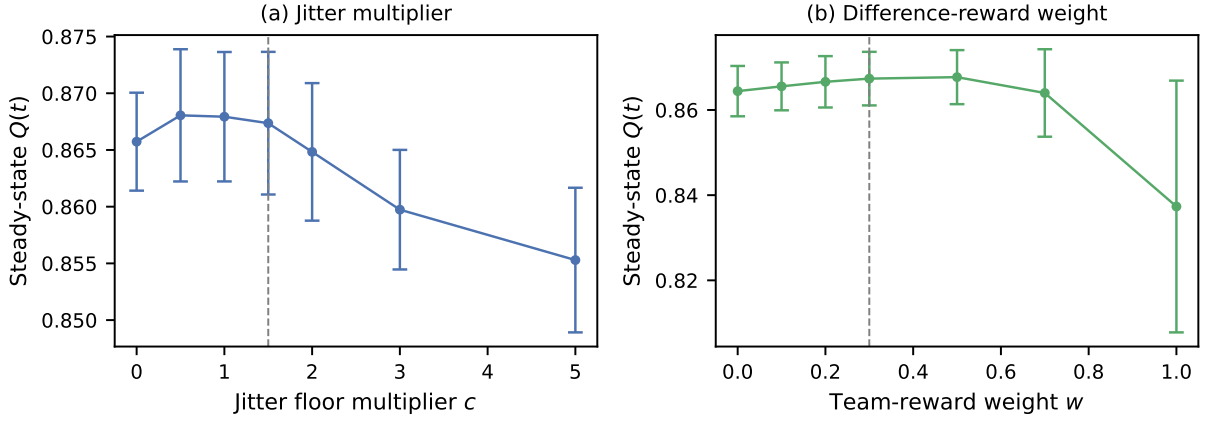


Fig. 4. Sensitivity of steady-state $Q(t)$ (mean $\pm$ one standard deviation, 5 seeds) to (a) the global-games jitter multiplier and (b) the difference-reward weight. Dashed lines mark the values used elsewhere in this paper.

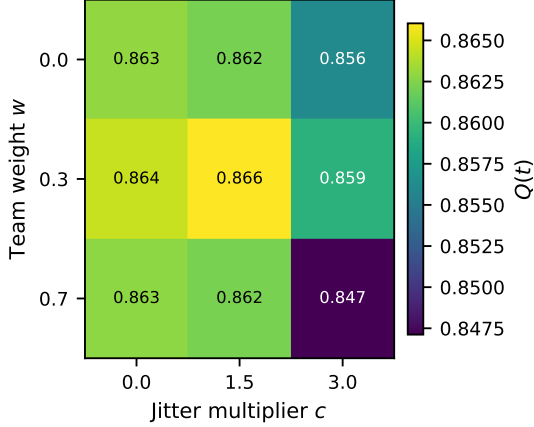


Fig. 5. Joint sensitivity of steady-state $Q(t)$ to jitter multiplier and difference-reward weight (3 seeds per cell). The performance penalty for elevated team weight grows with jitter level, indicating a genuine interaction rather than two independent effects.

action distribution to distinguish stable compliance from coordinated stampede.

7.2 Principle 2 — The Aggregation Operator for Multi-Factor Coordination Indices Must Preserve Comparability Across Revisions

Formal statement. When a coordination index is defined as a function of n normalized component scores, the aggregation operator must produce values that are comparable in magnitude across revisions of the index that introduce or retire components, and must not penalize the addition of a well-performing component by mechanically depressing the composite score.

Derivation from this study. A raw product of n sub-unity terms is not a mean: it lacks the $1/n$ normalizing root and therefore shrinks monotonically as n increases, independent of whether any component's value has worsened. This artifact was confirmed directly: four factors each valued at approximately 0.88 multiply to approximately 0.60—a

28-point drop from their arithmetic average—causing an apparent regression when S_4 was introduced despite no component having deteriorated. Replacing the raw product with the geometric mean preserves the logically desirable property that any factor approaching zero collapses the composite score (AND-gate semantics) while keeping the index magnitude comparable across versions of different dimensionality.

Generalization. Multi-factor health or quality indices in any adaptive system design—whether for scheduler coordination, network congestion control, or distributed consensus—should use the geometric mean or an equivalent power mean as the aggregation operator whenever the index is expected to evolve over time by the addition of diagnostic components.

7.3 Principle 3 — Closed-Loop Feedback Controllers Must Target the Sub-Metric Responsible for the Observed Deficit

Formal statement. A closed-loop controller in a signal-driven scheduling architecture converges only if the controlled variable has a direct causal path to the coordination sub-metric exhibiting the measured deficit. A controller that acts on a sub-metric already near its ceiling will saturate without measurable effect on the composite coordination index.

Derivation from this study. The original single-actuator PID controller was specified to retune the predictive layer's smoothing constant toward $Q \rightarrow 1$. In practice, the signal fidelity component S_1 was already high (mean 0.92), and the coordination deficit resided in S_2 (directive compliance) and S_3 (action coherence)—neither of which is causally connected to the smoothing constant. The controller saturated at the parameter's ceiling and remained pinned there for the entire simulation duration. The multi-actuator replacement decomposed the rolling deficit across sub-metrics and selected matched levers: per-agent perceptual jitter targeting S_4 , a shared switching penalty targeting S_3/S_4 , and a periodic Q-table consensus blend targeting S_3 . This design raised Q from 0.66 (Round 4) to 0.86 (Round 5).

Generalization. Any adaptive control loop for OS-level or distributed coordination should begin with an explicit

causal audit mapping each available actuator to the specific coordination sub-metric it can affect before the controller is implemented. Saturated actuators are silent failures: they produce no error signal and can persist undetected through conventional performance monitoring.

7.4 Principle 4 — Coupled Learning-and-Control Systems Require Explicit Two-Timescale Separation to Guarantee Convergence

Formal statement. When an outer closed-loop controller modifies the reward function or exploration parameters of an inner reinforcement-learning population, the coupled system does not satisfy the stationarity assumption underlying standard single-timescale convergence proofs. Convergence to a well-defined joint solution requires that the controller’s update magnitude vanish asymptotically faster than the learners’ update magnitude.

Derivation from this study. The multi-actuator controller of Section 5.3 updates the agents’ shared switching penalty—a parameter of the reward function the agents are simultaneously learning against. This coupling renders the Markov decision process non-stationary from the learners’ perspective. Two-timescale stochastic approximation theory [10], [11] provides the required convergence guarantee: when the outer loop’s step size decays strictly faster than the inner loop’s, the coupled system converges to a solution of an associated pair of ordinary differential equations. This is enforced in the implementation by restricting controller updates to once every 20 scheduling ticks and applying a step size decaying as $k^{-1/2}$ on the controller’s own update counter.

Generalization. Any signal-driven OS design that combines per-process reinforcement learning with an outer adaptive controller—whether for scheduling, memory management, or power governance—must explicitly implement and document the two-timescale separation. The absence of this separation does not merely slow convergence; it removes the theoretical guarantee that the system converges to any stable operating point at all.

7.5 Principle 5 — Reward Functions and Directive Functions in Learning-Based Schedulers Must Be Internally Consistent

Formal statement. In a signal-driven scheduler where process behavior is shaped by reinforcement learning, the reward function and the directive function that defines coordinated behavior must be jointly consistent: every action the reward function encourages must agree with the action the directive function specifies for the same system state, and vice versa.

Derivation from this study. Inspection of the original reward function against the directive function revealed a direct contradiction: the reward assigned positive value to the action RUN for CPU utilization $u_t \in [0.3, 0.7]$, while the directive specifies MIGRATE for $u_t > 0.6$. Agents were therefore being correctly trained toward a target that conflicted with the metric used to evaluate them—a specification error, not an exploration–exploitation tradeoff. Replacing the reward with a directive-match signal raised mean steady-state S_2 from 0.57 to 0.79 in a single round.

Generalization. In any OS or distributed system design where multiple adaptive components—learned policies, feedback controllers, and coordination metrics—are specified independently, a formal consistency audit should be conducted prior to simulation or implementation. Inconsistencies of this type do not produce immediate visible failures; they impose a silent ceiling on achievable coordination that is indistinguishable from a hard algorithmic limitation without source-level inspection.

7.6 Principle 6 — State Representations for Learning Agents Must Be Aligned With the Decision Boundaries of the Target Policy

Formal statement. The discretization of the state space for a tabular reinforcement-learning agent must be aligned with the decision boundaries of the policy the agent is being trained to approximate. Any discretization that merges states with different correct actions imposes a hard ceiling on achievable compliance that is independent of training duration and cannot be resolved by tuning learning-rate or exploration parameters.

Derivation from this study. A uniform four-bin discretization of CPU utilization placed states with directive RUN (e.g., $u_t = 0.60$) and states with directive MIGRATE (e.g., $u_t = 0.65$) in the same bucket. Because a tabular Q-learner assigns a single preferred action per state, this created an irresolvable ceiling on S_2 . Re-aligning bin boundaries to the directive function’s own decision thresholds raised mean steady-state S_2 from 0.79 to 0.90 without any other change.

Generalization. In any operating system design that combines discretized state abstractions with learned or rule-based response policies, the state representation must be validated against the target policy’s decision boundaries before training begins. This validation is a precondition for correct behavior and should be treated as a formal specification step, not an empirical tuning activity.

7.7 Principle 7 — Anti-Synchronization Mechanisms Should Be Derived From Equilibrium-Selection Theory Rather Than Empirically Tuned

Formal statement. In a shared-signal coordination game where multiple equilibria exist, including a pathological synchronized equilibrium, the magnitude of any private noise introduced to break equilibrium multiplicity should be derived from the signal’s intrinsic observation-noise standard deviation using global-games theory, rather than discovered through empirical tuning by an adaptive controller.

Derivation from this study. Global-games theory [12], [13] establishes that perturbing each agent’s observation of a common signal with idiosyncratic private noise eliminates equilibrium multiplicity in the small-noise limit, selecting a unique threshold equilibrium. The signal’s intrinsic observation-noise standard deviation σ_{obs} was estimated directly from a held-out calibration trace using a robust estimator (median absolute deviation of first differences [15]), chosen for resilience against load-spike outliers. Against a known injected noise of 0.030, the estimator recovered 0.0297 from a 500-tick trace. Initializing the jitter floor at $c\sigma_{\text{obs}}$ with $c = 1.5$ eliminated the need for the outer controller to discover a useful jitter magnitude from zero

through slow reactive feedback, reducing the risk of early-phase coordination collapse.

Generalization. Wherever a population of processes or agents shares a public predictive signal, the risk of synchronized equilibria should be addressed by principled noise injection scaled to the signal's natural noise floor, not by controller-discovered parameters. This approach provides a theoretically grounded floor for anti-synchronization noise, ensures the mechanism is proportionate to signal statistics, and reduces the number of free parameters requiring empirical justification.

7.8 Principle 8 — Individual Learning Agents Must Receive Causal Credit for Their Contribution to Population-Level Coordination

Formal statement. When a population of independently learning agents shares a global reward signal (or coordination index), each agent's learning signal is diluted by the joint actions of all other agents. A necessary condition for individual agents to develop behaviors that improve population-level coordination is that each agent receives a reward signal that reflects its counterfactual causal contribution to the population-level utility—its difference reward.

Derivation from this study. Prior to Round 6, all coordination pressure was imposed externally through jitter and Q-table consensus blending; no individual agent's reward reflected its own causal contribution to S_3 or S_4 . The difference reward (5) was introduced as a blend with the local directive-match signal at weight $w = 0.3$, following the formulation of Wolpert and Tumer [14]. A sensitivity analysis across five seeds confirmed that $w = 0.3$ maximizes the mean coordination index across the tested range and that removing the local directive signal entirely ($w = 1.0$) destabilizes learning, increasing cross-seed variance of $Q(t)$ nearly fivefold.

Generalization. Multi-agent scheduling architectures in which processes learn independently over shared system state should incorporate difference-reward credit assignment as a first-class design element, not as a post-hoc correction. Without it, coordination pressure must be entirely externally imposed, making the system brittle to changes in population size, workload heterogeneity, or signal characteristics.

7.9 Principle 9 — Adaptive Predictors Should Use Offline-Calibrated Parameters When Online Noise-Covariance Estimation Is Not Identifiable

Formal statement. For a local-level state-space model of system load, the ratio of process to observation noise is the single parameter governing steady-state Kalman gain. This ratio is not identifiable from lag-zero innovation statistics alone, and attempts at online estimation from the innovation sequence will misattribute genuine load spikes to observation noise, causing prolonged periods of filter overconfidence. Offline calibration of the noise-ratio parameter using a held-out trace is both more reliable and more general than online adaptive estimation.

Derivation from this study. An online-adaptive Kalman filter was implemented and evaluated against a 3,000-tick evaluation trace. Its mean squared forecast error was

0.0286—approximately seven times worse than a naive fixed-parameter exponential smoother (MSE 0.0041–0.0057). Diagnostic analysis confirmed the cause as the classical noise-identifiability problem [9]: when a genuine load spike arrived while the filter's estimated uncertainty was low, the entire innovation was attributed to observation noise, collapsing the Kalman gain and causing the filter to disregard real signal data for over 100 subsequent ticks. An offline-calibrated estimator, fitting the single noise-ratio parameter via one-dimensional grid search on a separate calibration trace, achieved MSE 0.00404—matching the best hand-tuned fixed smoother exactly (MSE 0.00406) while remaining generalizable to different load statistics.

Generalization. Operating systems and schedulers incorporating Kalman-filter-based prediction of load, thermal, or memory-pressure signals should default to offline calibration of filter parameters unless a provably identifiable observation model is available. The failure mode of online adaptive Kalman filtering documented here is not implementation-specific; it is a structural consequence of the identifiability limitation and will manifest in any system where load spikes are large relative to baseline variability.

7.10 Principle 10 — Simulation-Driven Validation Is a Necessary Intermediate Stage Between Architectural Design and Kernel Implementation

Formal statement. Pre-implementation userspace simulation is not merely an accelerated prototype; it is a qualitatively distinct validation stage capable of revealing failures in metric design, control architecture, learning specification, and theoretical consistency that are not detectable through inspection of the architectural description alone and are substantially more costly to discover in a running kernel.

Derivation from this study. Seven of the failures diagnosed and repaired in this study were invisible in the architectural description but surfaced immediately under simulation: (i) a metric blind spot allowing perfect scores under stampede behavior; (ii) a controller saturating without effect; (iii) a reward-directive contradiction; (iv) a state discretization misaligned with decision boundaries; (v) a missing exploration schedule imposing a structural compliance ceiling; (vi) an identifiability failure in the adaptive predictor; and (vii) absent individual-level incentives for coordination. Each was discovered by observing a concrete, reproducible quantitative failure—not by re-reading the specification. The iterative structure—simulate, diagnose a specific failure, apply the most principled available fix, re-measure—is itself a reusable methodology applicable to any coordination-based OS design.

Generalization. Future signal-driven operating systems incorporating predictive signal buses, decentralized adaptive scheduling, reinforcement-learning-based response functions, and closed-loop coordination controllers should treat simulation-driven validation as a mandatory gate between architectural specification and kernel-level implementation. The cost reduction comes not only from cheaper bug discovery but from the ability to apply theoretical tools—stochastic approximation analysis, equilibrium-selection theory, multi-agent credit assignment—in a controlled, observable environment before hardware constraints limit instrumentation.

TABLE 3
Summary of the ten design principles for signal-driven operating systems.

	Core Claim	Primary Evidence
P1	Coordination metrics must encode temporal stability, not only instantaneous compliance	Introduction of S_4 ; baseline score corrected from 1.00 to 0.86
P2	Multi-factor indices must use the geometric mean for cross-version comparability	Raw-product artifact: four 0.88-valued factors yielding 0.60
P3	Feedback controllers must act on the sub-metric causing the deficit	PID saturation; multi-actuator redesign raising Q by 0.20
P4	Coupled learning-and-control systems require explicit two-timescale separation	Borkar [10], [11]; enforced via 20-tick controller cadence and $k^{-1/2}$ decay
P5	Reward and directive functions must be jointly consistent	Reward-directive contradiction; single fix raises S_2 from 0.57 to 0.79
P6	State discretization must align with policy decision boundaries	Uniform binning imposing structural S_2 ceiling; re-alignment raises S_2 to 0.90
P7	Anti-synchronization noise should be derived from equilibrium-selection theory	Global-games sizing; $\hat{\sigma}_{\text{obs}} = 0.0297$ vs. known 0.030
P8	Individual agents must receive causal credit for population-level coordination	Difference rewards; $w = 1.0$ fivefold variance increase confirms necessity of blending
P9	Predictive filters should use offline calibration when online estimation is not identifiable	Online Kalman MSE 0.0286 vs. offline 0.00404; Mehra [9] identifiability limit
P10	Simulation-driven validation is a necessary gate before kernel implementation	Seven failures invisible in specification but reproducibly observable in simulation

8 DISCUSSION AND LIMITATIONS

The simulation reported here is deliberately restricted to a single scheduling tier (core-local signal aggregation); node- and cluster-level tiers, inter-node clock synchronization, and network propagation delay for the shared signal remain unmodeled and are appropriate targets for a subsequent simulation phase before any claims are made about multi-node behavior. The cryptographic verification model assumes a verifier already possesses the correct rotating key; key distribution across cores and nodes is a separate problem not addressed here. The learning agents are tabular Q-learners chosen for implementation speed and interpretability; the underlying architecture does not mandate a particular learning algorithm, and the state-discretization ceiling identified in this work (Section 5.5) would not necessarily manifest under a function-approximation-based policy, though a different set of failure modes might. Finally, all quantitative results in this paper derive from a single synthetic hardware-load trace family; the calibration and jitter-sizing procedures are designed to transfer to other trace statistics automatically, but this transfer has not been empirically verified against a qualitatively different load pattern. Additionally, the round-by-round progression reported in Table 1 derives from a single random seed (seed 42); the multi-seed sensitivity analysis of Section 6.4 validates the final configuration across five seeds but does not extend to earlier rounds, and the trajectory of $Q(t)$ across rounds should therefore be interpreted as illustrative rather than statistically representative.

9 CONCLUSION

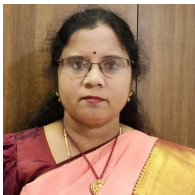
We presented ORCHESTRA-OS, a predictive, cryptographically secured, hierarchically signal-coordinated scheduling architecture, together with a simulation-based validation methodology applied through seven iterative rounds, each triggered by a specific, reproducible failure. Five of the seven rounds addressed conventional software or reinforcement learning engineering defects (a reward function con-

tradicting its own scoring metric, a discretization scheme misaligned with decision boundaries, missing exploration annealing, a metric blind spot, and a controller acting on the wrong variable); two rounds applied established mathematical theory—global games and difference rewards—to replace empirically-tuned heuristics with derived, validated quantities. The resulting system exceeds a structurally advantaged reactive baseline that copies rather than infers the target policy, establishing that signal-driven coordination through learned adaptive response functions is computationally achievable at the userspace simulation stage; a multi-seed sensitivity analysis confirms the two remaining free parameters lie in robust, low-variance regions of the design space rather than at narrow optima requiring further tuning. The ten design principles distilled in Section 7 generalize the study’s findings beyond ORCHESTRA-OS to any parallel or distributed system that coordinates adaptive clients through a shared predictive signal, and the seven-round structure—simulate, diagnose a specific failure, apply the most principled available fix, and re-measure—is offered as a template for validating comparable coordination-based system designs prior to committing to kernel-level or other high-cost implementation. More generally, this study suggests that simulation-driven architectural validation can serve as a rigorous scientific methodology for developing coordination-based operating systems—one in which simulation functions not merely as verification but as a process of discovery.

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